

“A Review of Magnetic-Field Sensing Technologies: Applications in Energy, Transportation & Industry”

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Abstract (199 words):

Magnetic-field sensors enable non-invasive measurement of electrical current and magnetic fields generated by energy, transportation, and industrial systems, such as grid transformers, automotive drivetrains, and robotic motion-tracking systems. These sensors assess mechanical motion, material characteristics, and equipment condition for prediction, control, and forecasting, improving reliability and efficiency while reducing costs. A clear understanding of sensor operating principles, transduction mechanisms, and operational limits helps identify suitable applications and clarify each technology's technical value. In this review, we compare established and emerging magnetic-field sensing technologies and their integration with new artificial intelligence (AI) tools, relating performance and commercial maturity to representative applications across these domains. In addition, our review examines the implementation of data-driven methods in non-invasive magnetic-field sensing for real-time and in situ measurement, digital twins, and smart, adaptive, and reliable operation under realistic and dynamic conditions. This advanced sensing-system approach fully exploits each technology's strengths and improves measurement quality, system capability, and decision-making. Future progress will depend on integrating sensing hardware and computation while supporting traceability and compliance with relevant industry standards and government policy. By connecting sensing principles with application value and technology readiness, this detailed review provides a practical framework for magnetic-field sensing technology development and research.